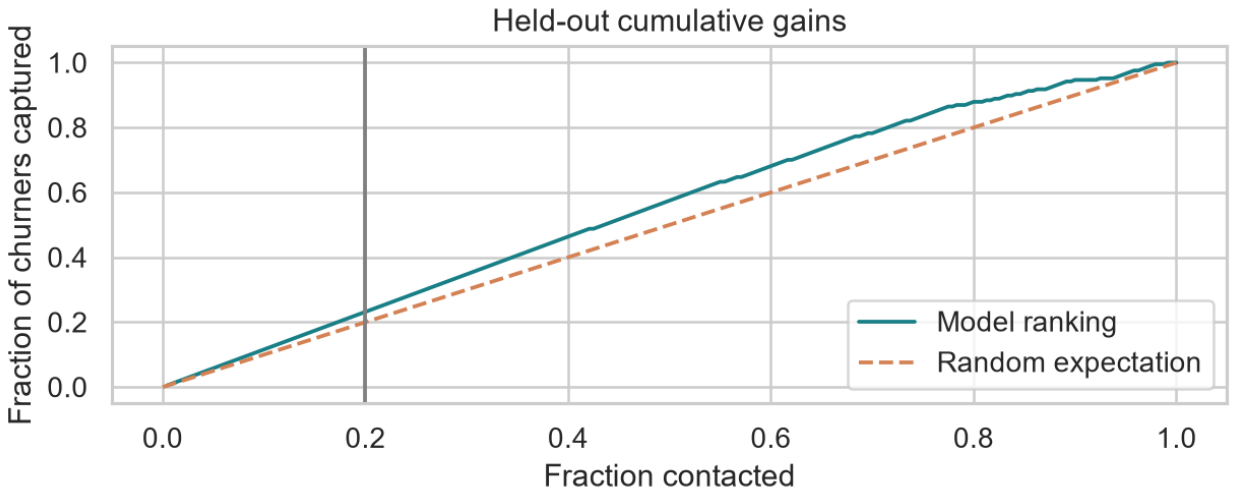
01 / Customer retention analytics

DECISION BRIEF | Subscription services | Reproducible analysis | 21 October 2026

The selected Boosting model achieved held-out ROC-AUC 0.880 and average precision 0.979. The AP random-ranking benchmark is 0.863; high prevalence makes apparently high AP and accuracy easier to obtain.

Decision measure	Held-out result
Contact capacity	48 / 240 customers
Churners captured	48; random expectation 41.4
Precision / recall at 20%	100.0% / 23.2%
Lift over random	1.159x
Hypothetical incremental net value	$330



Use the ranking as a pilot outreach queue. Prediction identifies association, while incremental retention value requires a controlled experiment. Treat the unusually high churn prevalence as a sampling and target-definition validation priority.

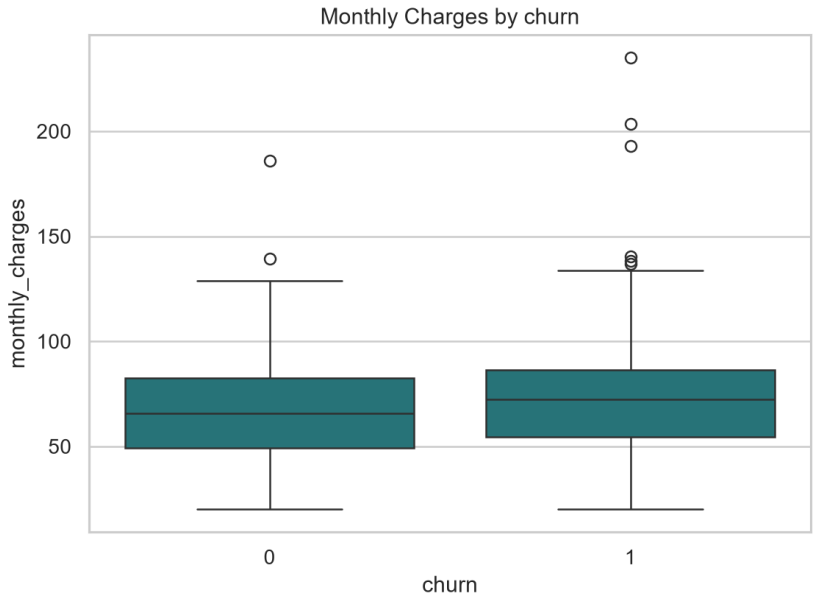
02 / Data quality and preparation

Source: customer_churn_advanced_dataset.csv. 1,200 customers and 20 columns; 1,036 churn labels (86.33%) and 164 non-churn labels (13.67%). customer_id uniquely identifies every record. No exact duplicate rows were found.

Missing predictor	Count	Percent
monthly_charges	30	2.5%
total_charges	24	2.0%
satisfaction_score	48	4.0%
monthly_usage_gb	36	3.0%

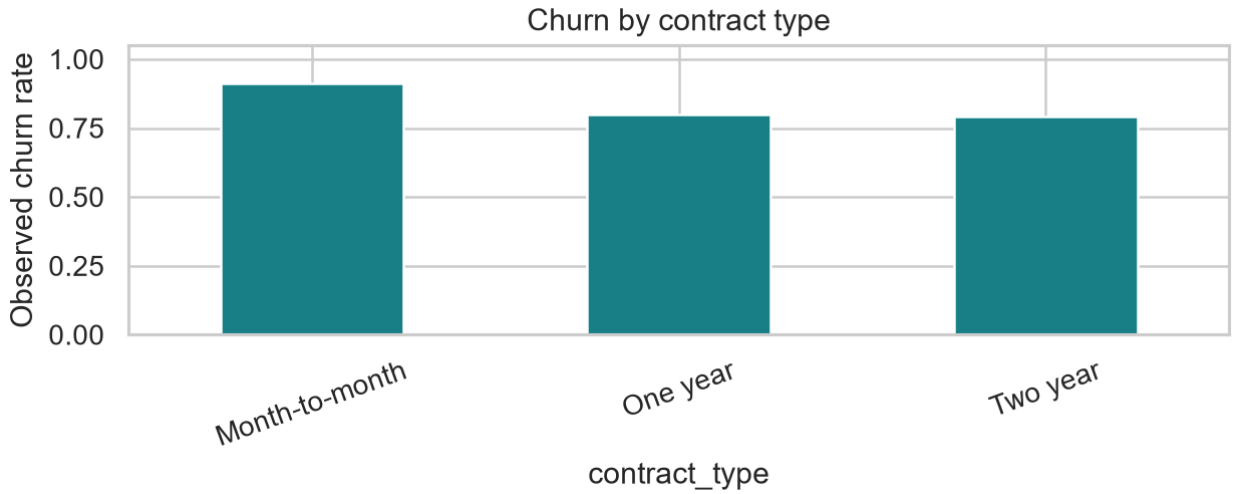
The validated cleaned export preserves all records, identity, target labels and missing values. No full-dataset imputation is performed. Numeric medians and categorical modes are learned in each training fold. Scaling and unknown-safe one-hot encoding are contained in the pipeline.

Monthly charges range from 20 to 235.01; usage ranges from 2 to 466.90 GB. Extremes and historical/current billing ratios are explicitly exported for investigation. Current prices times tenure need not equal total historical charges because plans, discounts and prices may change. Without invoices, these records cannot be classified as errors, so no arbitrary capping or deletion is applied.

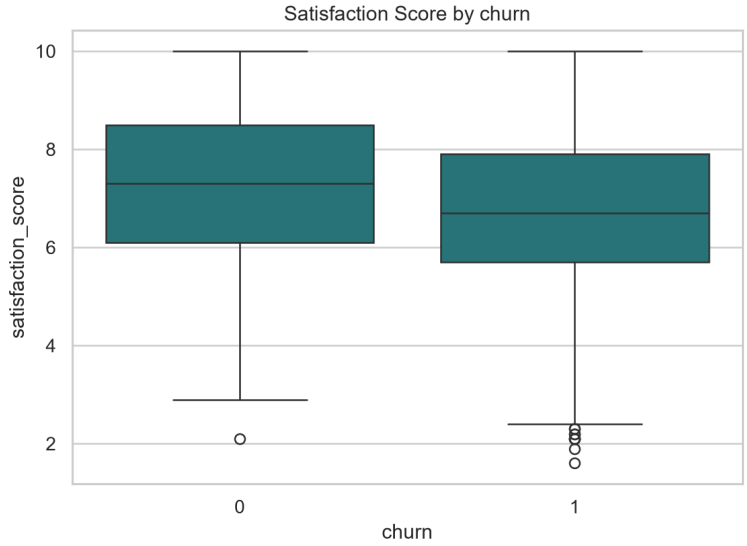


Domain checks cover age, tenure, nonnegative financial/behavioral quantities, satisfaction 1-10, discounts 0-100 and binary senior status. Observed category vocabularies are documented in the notebook; no authoritative external vocabulary was supplied.

03 / Observed churn patterns



contract_type: highest observed rate Month-to-month (91.1%); lowest Two year (79.1%). Group counts above qualify these descriptive associations; they do not establish intervention effects.



Median satisfaction_score: non-churn 7.30, churn 6.70. Overlap and extreme observations limit simple cutoffs; missing observations are omitted in this chart.

The notebook also analyzes tenure, complaints, late payments, internet service, monthly usage, discounts, support, payment method, missingness and interactions. All exploratory outcome summaries use training customers only. Small subgroups and overlapping distributions limit interpretation.

04 / Features and model selection

A stratified 80/20 split locks 240 test customers before EDA. Five shuffled stratified folds on the 960 training customers evaluate all baseline and tuned candidates, with seed 42. Average precision selects the model; ROC-AUC, recall, F1 and top-20 precision qualify that decision.

Eight reusable features represent lifecycle bands, monthly charge per tenure, complaint and late-payment exposure over min(tenure,12), inverse satisfaction, a service/support gap, usage per charge, and a dissatisfaction/complaint interaction. Safe denominators and missingness handling are implemented in a serializable transformer. ID and target are excluded by an explicit feature allowlist.

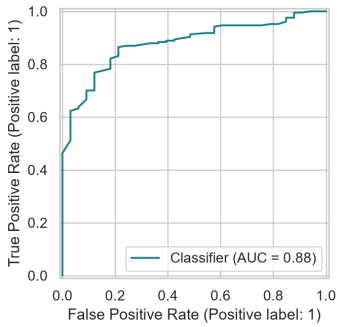
Model / version	CV AP mean (SD)	CV ROC	Top20 precision
Dummy / baseline	0.864 (0.002)	0.500	0.872
Logistic / baseline	0.970 (0.017)	0.860	0.979
Logistic / tuned	0.971 (0.014)	0.867	0.979
Forest / baseline	0.968 (0.010)	0.854	0.979
Forest / tuned	0.972 (0.010)	0.867	0.985
Boosting / baseline	0.970 (0.008)	0.866	0.985
Boosting / tuned	0.977 (0.005)	0.878	0.990

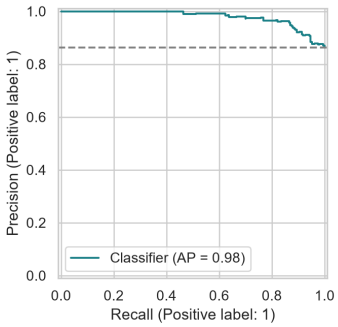
Search spaces: Logistic C={0.1,1,10}, weights={none,balanced}; Forest depth={4,unlimited}, minimum leaf={2,8}, weights={none,balanced}, 120 trees; Boosting trees={60,120}, rate={0.03,0.1}, depth={1,2}. Every candidate uses five folds. Search results and training feature-ablation results are in the executed notebook.

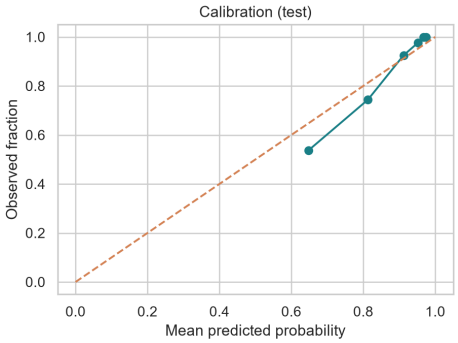
Selected family: Boosting; engineered features retained: True. Features are selected by training CV. Fold dispersion supplies a robustness check. Winning CV estimates have search-selection optimism; they are distinct from the final held-out evaluation.

05 / Evaluation and probability reliability

Metric	Test value
accuracy	0.8625
precision	0.8625
recall	1.0000
f1	0.9262
roc_auc	0.8798
average_precision	0.9791
brier	0.0900





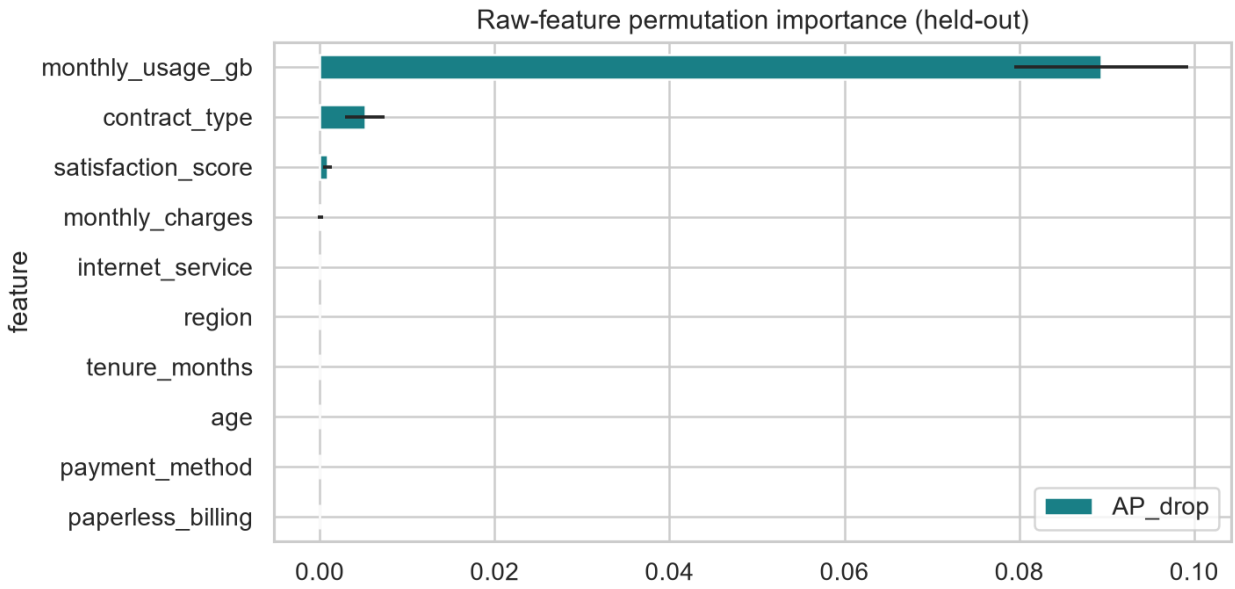


The fixed classification threshold is 0.410, selected from training out-of-fold predictions by hypothetical net utility: $200 retained value, 25% success and $5 per contact. This unconstrained classifier is separate from the fixed 20% contact queue. False negatives lose potential retained value; false positives consume contact cost.

Calibration is checked using quantile bins and Brier score, alongside a constant-prevalence benchmark in the notebook. Few non-churn cases make reliability estimates imprecise. No post-test calibration or threshold changes were made. Accuracy alone is insufficient because predicting churn for everyone already yields approximately 86% accuracy.

Robustness: the CV comparison includes fold-to-fold standard deviations and a training-only feature ablation. One split, small minority class, missing timestamps and uncertain measurement timing prevent claims of temporal or external validity. Non-nested OOF threshold utility can retain model-selection optimism.

06 / Risk explanations and customer evidence



Global evidence is held-out raw-feature permutation AP loss, repeated eight times. Raw-column permutation aggregates encoded levels and propagates through derived features. Correlated predictors can share importance. These results are never used for feature selection.

Low risk: CUST100626, probability 45.6%. Strongest reference replacements: monthly_usage_gb: -0.406; contract_type: -0.129; satisfaction_score: -0.057.

Medium risk: CUST100019, probability 94.3%. Strongest reference replacements: monthly_usage_gb: +0.084; contract_type: -0.022; satisfaction_score: -0.011.

High risk: CUST100178, probability 97.5%. Strongest reference replacements: monthly_usage_gb: +0.041; satisfaction_score: +0.002; monthly_charges: +0.000.

Each delta is the original probability minus the score after replacing that raw attribute by its training median/mode, recalculating derived features. The notebook includes the actual records. This is a model-sensitivity explanation, not additive attribution or a causal treatment effect. Reference replacements can create implausible combinations. Medium denotes the cohort median score rank.

07 / Retention economics and operational next steps

Contact 48 customers, reaching 48 observed churners. Under the base assumptions, expected retained customers = 12.00; cost = $240; gross value = $2400; net benefit = $2160. Incremental net value over uniform random outreach = $330.

Success	Value	Model net	Random net	Difference
10%	$100	$240	$174	$66
10%	$200	$720	$588	$132
10%	$400	$1680	$1416	$264
25%	$100	$960	$795	$165
25%	$200	$2160	$1830	$330
25%	$400	$4560	$3900	$660
40%	$100	$1680	$1416	$264
40%	$200	$3600	$3072	$528
40%	$400	$7440	$6384	$1056

Pilot the top-risk queue: contact 48 of 240 comparable customers; measured capture is 48 churners versus 41.4 expected randomly.

Use the contract-rate evidence to test renewal reminders and contract-fit conversations; measure incremental retention by randomized assignment.

Use satisfaction and complaint evidence to triage service recovery; test response quality and timing rather than assuming a causal benefit.

Use late-payment and payment-method evidence to test opt-in billing assistance; avoid punitive targeting or sensitive financial profiling.

Use support and usage evidence to test relevant support onboarding and plan-fit reviews. Audit subgroup access and outcomes, including age and region.

Assumptions are editable in CFG. These figures are hypothetical, not experimental ROI. Validate target provenance, scoring-time availability, temporal generalization, consent, fairness across age/region, offer costs and heterogeneous treatment effects. Run a randomized retention experiment; monitor calibration, data drift and customer burden before expansion.